MIL-W-60306A
28 January 1969
SUPERSEDING
MIL-W-60306(MU)
19 October 1967

MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, SMOKE, M156;

WP FILLING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the filling of a 2.75 inch rocket head with white phosphorus (WP).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specifiction to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-P-45449 - Pallet, Units, Wood, for Shipment of Projectile Metal Parts, and Projectile Ammunition.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" (Partial) Pallet Skids, Runners, or Pallet-Type Base.

FSC 1340

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL90-1-44 - Warhead, 2.75 Inch Rocket, Smoke, WP, M156.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing DL90-1-44 and subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.2 Assembly. The warhead shall be assembled in accordance with Drawing D90-1-43.

3.3 Filling. The warhead, clean and free from foreign matter, shall be filled with molten WP. When water is in contact with the WP during the filling operation, the water used shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. The depth of water remaining on the surface of the WP in the filled and closed warhead shall not exceed 1/8 inch. The filling shall be solidified with the warhead in a nose-up (filling end) position after inspection in accordance with 4.2.4.1. (See 6.2 and 6.3.)

3.4 Filling height. The void remaining above the WP and water (when applicable) in the filled and closed warhead shall be 3 to 5 percent by volume. The head shall be in a nose-up position and the WP at a temperature of 130° ± 10°F (see 6.3), when examined in accordance with 4.2.4.1.

3.5 Cleaning after closing. The exterior surface of the filled and closed warhead and the interior of the burster casing shall be thoroughly cleaned to remove any grease, oil, WP, or other foreign matter (see 6.4).

3.6 Leakage. The warhead shall not leak when maintained at a temperature of 207° to 212°F for a minimum period of 15 minutes or 160 + 5°F for a minimum period of 8 hours, when examined in accordance with 4.2.4.2. After the warhead has been accepted, the WP shall be allowed to solidify in a nose-up position.

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3.7 Workmanship. The warhead shall be free from contamination and damage such as cracks or deformation which could affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the warheads of one interfix number, filled by one manufacturer, at one plant, from the same materials under essentially the same manufacturing conditions. However, no more than one interfix lot of warhead metal components from one manufacturer shall be represented in any one lot of filled warheads.

4.2.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 Examination and nondestructive tests. Sample filled warheads and the preparation for delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 Critical defects. Each filled warhead in the lot shall be inspected for critical characteristics listed in the classification of defects.

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4.2.3.3 Classification of defects.

(a) Head and casing assembly (Drawing D90-1-43).

Categories	Defects	Acceptance standards
Critical:		
1	WP contamination (3.5)	
2	Leakage	4.2.4.2
Major:	AQL 0.65 percent defective	
*101	Filling excessive	4.2.4.1
	AQL 1.0 percent defective	
102	Burster casing location incorrect	
103	Foreign matter in burster casing	
104	Identification marking incorrect or illegible	
*105	Water overlay excessive	4.2.4.1
Minor:	AQL 2.5 percent defective	
*201	Filling insufficient	4.2.4.1
202	Adapter thread protector missing	

*Note: Sampling for inspection for these characteristics, after first acceptable lot, shall be in accordance with MIL-STD-105, level 1.

(b) Preparation for delivery (section 5).

Categoires	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Packing method incorrect
102	Strapping missing
103	Strapping loose
104	Lot marking incorrect or illegible
105	Marking incorrect or illegible

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4.2.4 Tests.

4.2.4.1 Filling level. The height of the water overlay and of the WP in the filled warhead shall be determined radiographically while the WP is still molten. The warhead shall be oriented in a vertical nose-up position.

4.2.4.2 Leakage. Following the final operation, the filled and closed warhead shall be placed in an oven in a nose-up position for test temperature of 207°F, and in a nose-down position for temperature of 160°F, and subjected to the required temperature for the specified period. The emission of smoke constitutes leakage.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and preservation. Warheads shall be clean and free of corrosion, and critical surfaces shall be preserved in accordance with applicable drawings and MIL-P-116.

5.2 Packing, level C. The filled warheads shall be palletized in a nose-up position in accordance with grade B, type IIa, style optional, specification MIL-P-45449. The cover assembly shall be banded securely to the pallet deck using steel strapping in accordance with MIL-STD-147.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Caution. Due to the hazardous nature of WP, filling installations are advised to take all possible precautions to protect personnel from injury while filling.

6.3 Filling aid. Warheads filled with approximately 2 pounds of WP before closing have been found to comply with 3.4. This information is furnished as an aid to procurement and not as a criterion for filling.

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6.4 Decontamination. The filled warhead may be cleaned by passing through a hot water and detergent spray washer or by flushing with hot water while vigorously scrubbing with steel wool. Additional cleaning of the burster well may be effected with a steam lance and a power driven wire brush.

Custodians:

Army - MU
Air Force - 70

Review activity:

Air Force - 70

User activities:

Navy - AS
Air Force - 15

Preparing activity:

Army - MU(EA)

Project No. 1340-0046

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF [illegible] WHICH MAY BE USED

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OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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